Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

2 SQA 2025, question 14 (continued)

- (b) As the temperature rises the thermistor's resistance falls (TURD), so V_{in} falls. Below about 0.7 V the transistor switches off, so no current flows from collector to emitter and the LED switches off.
- (c) The resistor limits the base current, protecting the transistor.
- (d) $V_2 = 4800 \div 3500 \times 2.1 = 2.88$ V; $V_s = 2.1 + 2.88 = 5.0$ V (or $I = 2.1 \div 3500 = 0.0006$ A, $V_2 = 0.0006 \times 4800 = 2.88$ V)

4 SQA 2024, question 8 (continued)

- Parallel section: $(390 \times 270) \div 660 = 160\ \Omega$
- Total: $780 + 160 + 1200 = 2140\ \Omega = 2.1\ k\Omega$
- Current: $I = 1.2 \div 390 = 0.0031$ A (3.1 mA)
- Variable resistance increased: the voltage across the other components decreases; the current decreases (total resistance rises)
- Simulation: components aren't destroyed, so testing is cheaper; components are quick to change; software highlights faults; it is safer

7 SQA 2024, question 10 (continued)

- Thermistor resistance from the graph: 800 Ω
- Name: voltage (potential) divider
- $V_1 = 5 - 1.4 = 3.6$ V; $R = 1.4 \div 3.6 \times 4800 = 1.9\ k\Omega$
- As temperature rises the thermistor's resistance falls, so V_{in} rises. At about 0.7 V the transistor switches on, the relay activates, the blue LED switches off and the red LED switches on.

9 SQA 2023, question 10(d) (continued)

- $1/R_p = 1/82 + 1/78 + 1/86 = 0.0366$, so $R_p = 27\ \Omega$
- Total: $390 + 27 = 420\ \Omega$
- $I = P \div V = 1.8 \div 12 = 0.15$ A

12 SQA 2023, question 14 (continued)

- As temperature rises the thermistor's resistance falls, so V_{in} rises. At about 0.7 V the transistor switches on and the relay activates: the red LED switches off and the green LED switches on.
- Thermistor: $V_1 = 5 - 1.9 = 3.1$ V; $R = 3.1 \div 1.9 \times 1700 = 2.8\ k\Omega$
- When the relay switches: the red LED switches off and the green LED switches on
- Simulation: quicker, no components wasted, easy to change, no risk to the user
- A compound gear train uses smaller gears, so it fits inside the small laminator